

3.1 Introduction

A need remains to create an accessible and effective solution for rural farmers. Despite the creation of IoT technologies that monitor soil moisture, existing systems still lack informativeness. With informative usability, not only can the farmer easily read and understand the real-time data, but they can also receive actionable recommendations on what to do with that information. This chapter covers various research methods, including the design, development, and testing of the proposed system. Additionally, this chapter provides a comprehensive explanation of the processes involved in developing and validating the suggested prototype using a design research methodology.

3.2 Research Methods

To develop a highly accessible and effective solution, this research utilizes the Design Science Research Methodology (DSRM) (Vom Brocke *et al.*,2020^{xi}). This methodology is particularly well-suited for the creation and improvement of prototypes that respond to real-world challenges, making it a perfect fit for the objectives of this study. DSRM is characterized by its iterative stages, which include identifying the problem, designing the solution, developing the prototype, evaluating its effectiveness, and refining it based on feedback. Through this structured approach, the research aspires to create a user-centred and functional system that enables farmers to make informed, data-driven decisions regarding their irrigation practices.

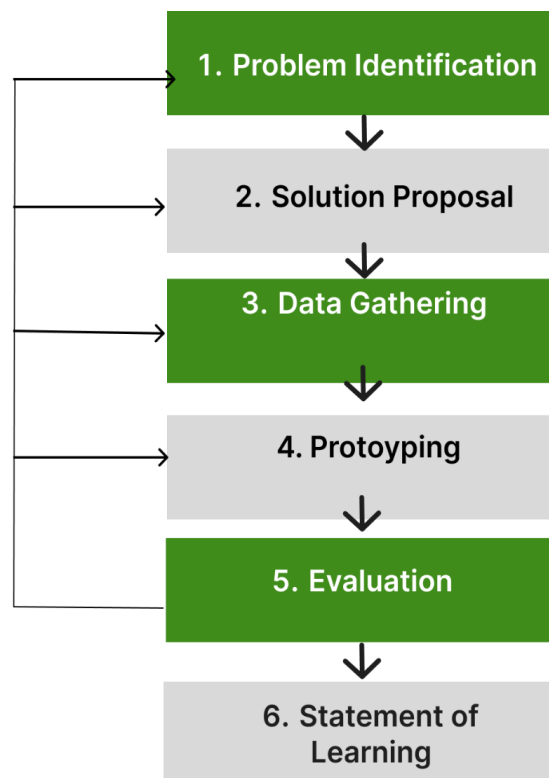


Figure 1: Design Research Process

As depicted in Figure 1, this approach aims to ensure that the prototype is effective and userfriendly, contributing to better water management and improved agricultural productivity.

3.3 Problem Identification

Farmers today are grappling with a multitude of pressing challenges, particularly the scarcity of water, which severely impacts their agricultural output. This issue is exacerbated by the rising costs of irrigation, making it increasingly difficult for them to maintain their crops effectively. Additionally, many rural farmers lack access to advanced and affordable technology that could facilitate efficient water management. Compounding these difficulties is a shortage of resources and knowledge, which often results in outdated and ineffective irrigation practices.

While there are sophisticated soil moisture detection systems on the market that could greatly benefit farmers, their high prices and complexity render them inaccessible or impractical for many in rural areas. Recognizing these significant hurdles, this research focuses on developing a costeffective, Internet of Things (IoT)--enabled soil moisture monitoring system. This innovative solution aims to empower farmers by providing them with easy access to actionable data, ultimately enhancing their ability to manage water resources effectively and sustainably.

3.4 Solution Proposal

Before smart technologies evolved, various approaches were used to detect the soil content, and most of these approaches were operated manually. Later on, smart agriculture was then introduced. The proposed prototype is an affordable IoT solution that leverages fog computing to help rural South African farmers manage irrigation efficiently. It includes soil moisture sensors connected to an Arduino microcontroller, which collects real-time data on soil conditions. This data is processed by a Raspberry Pi, acting as a fog node, to enable immediate, local data analysis. This setup minimizes latency, ensuring quick responses to irrigation needs. Important insights from the data are stored in the cloud for long-term analysis, allowing farmers to track soil moisture trends and gain a deeper understanding of seasonal crop requirements.

Farmers can access the system through a user-friendly web application, where they can create and manage profiles, view soil moisture analytics, and receive recommendations for optimized irrigation. The application features an AI-powered chatbot that retrieves processed data from the fog node and provides guidance in simple language, making advanced data insights accessible and actionable. This chatbot can answer questions, provide irrigation tips, and issue alerts when moisture levels are critically low. By prioritizing affordability, usability, and accessibility, the prototype empowers farmers to make data-driven irrigation decisions,

potentially increasing crop yields, reducing water waste, and promoting sustainable agricultural practices in resource-limited rural areas.

3.5 Data Gathering

This study has harnessed existing datasets that are readily available and sourced from various educational institutions, investigative organizations, and rural development bodies. Specifically, the primary dataset utilized for both the development and testing of the prototype was obtained from Kaggle, focusing on the Agricultural Data for Rajasthan from the years 2018 to 2019. This comprehensive dataset comprises 769 records, encompassing approximately 553.5 KB of information regarding agricultural water usage in the region of Rajasthan.

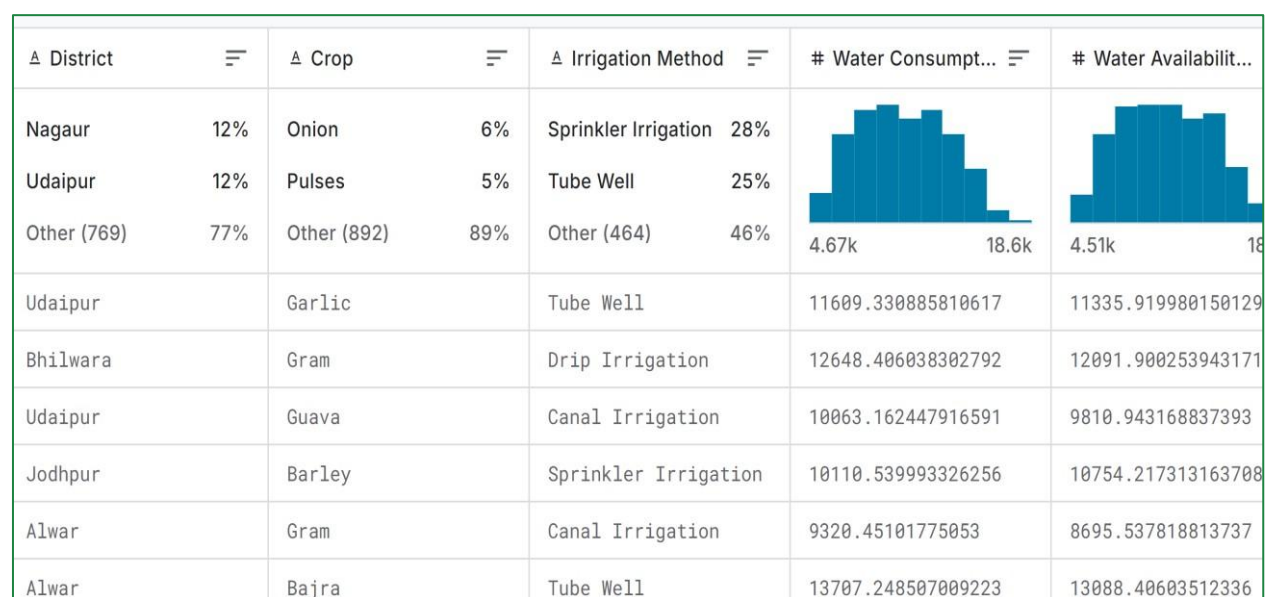


Figure 2: Summary of Water Usage from different Irrigation systems

DataSample:water_usage_data [Agricultural Data for Rajasthan, India (2018-2019) (kaggle.com)]

Data Size: 553.5 KB

Records: 769 records

Repository: Kaggle

URL: [Agricultural Data for Rajasthan, India \(2018-2019\)](https://www.kaggle.com/datasets/rajeshkumar1998/agricultural-data-for-rajasthan-india-2018-2019)

Figure 2 is particularly valuable as it offers detailed insights into the patterns of water consumption that vary across different locations in Rajasthan, allowing to adapt of the prototype to address specific local agricultural needs effectively.

3.6 Prototyping

Prototyping is a crucial phase in developing the soil moisture monitoring system. The design and development of the prototype utilize a comprehensive technology stack that supports both UI/UX design and backend functionalities. This prototype integrates IoT with fog computing to provide real-time soil moisture data, which is accessible through a user-friendly web application.

Design and Development Technology Stack:

- Front-end: PythonHTML, CSS, JavaScript
- Back-end: Python
- Database: Firebase
- Framework: Flask
- Sensors: Soil moisture sensors for real-time data collection
- Single Board Computers (SBC): Arduino Mega and Raspberry Pi 3 for local processing and data transmission.

3.7 Evaluation

Data analysis is transforming, cleaning, and modelling raw data to extract valuable insights and information. Its primary objective is to clarify complex data sets, uncover patterns and trends, and support informed decision-making (Das *et al.*,2023^{xii}).

One practical application of data analysis involves processing and visualizing real-time sensor data related to soil moisture. This approach generates actionable recommendations to optimize irrigation practices. By creating charts and graphs, stakeholders can visualize soil conditions, identify trends, and analyze irrigation patterns, ultimately validating the effectiveness and accuracy of the prototype.

The proposed prototype holds significant value, especially for small-scale farmers, due to its integration of advanced technology and cost-effectiveness. The cost-effectiveness of the prototype will be evaluated by comparing its implementation costs with the financial benefits it provides. To check usability, the study will gather feedback from farmers about how easy and effective the prototype is. If productivity increases due to better irrigation, then informative results will automatically improve usability.

3.8 Statement of Learning

To create an accessible interface for farmers, the system blends real-time data and valuable insights to optimize irrigation, thereby enhancing productivity and increasing cost-effectiveness. To achieve this, these components were used:

- **Dashboard:** This feature displays soil moisture information, providing farmers with realtime data or trends. This helps them make informed decisions about when to irrigate their fields.
- **Chatbot:** The chatbot facilitates informative interactions by answering farmers' queries related to soil health, moisture levels, and irrigation suggestions. It utilizes soil data to offer personalized guidance based on the specific needs of users.
- **Graphs (Dashboard):** The graphical representation shows the cost-effectiveness of optimized irrigation by illustrating data trends and the financial benefits gained. This helps users comprehend the economic impact of using the system in their farming practices.